Introduction to Gradient Descent

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Brief Review of Convexity

The materials is taken from

- Ahmadi (2015) lecture 7;
- Audet and Hare (2017) chapter 2;
- Boyd and Vandenberghe (2004) chapter 2;
- Schmidt (2018) lecture 3.

A convex optimization problem can be written in the form

$$\min_{x \in \Omega} f(x)$$

where f is a convex function and Ω is a convex set.

Definition 1 (Convex Set)**.** A set Ω is convex if and only if $\forall x, y \in \Omega$, and $\theta \in [0, 1]$, we have

$$\theta x + (1 - \theta) y \in \Omega$$

Definition 2 (Convex Combination)**.** A point $z \in \mathbb{R}^n$ is a convex combination of the points in $\{x^{(1)}, \ldots, x^{(m)}\} \subseteq \mathbb{R}^n$ if there exists scalars $\theta^{(1)}, \ldots, \theta^{(m)} \geq 0$ such that

$$z = \sum_{i=1}^{m} \theta^{(i)} x^{(i)} \quad \text{with} \quad \sum_{i=1}^{m} \theta^{(i)} = 1$$

Corollary 1. A set Ω is convex if convex combinations of points in the set are also in the set.

Definition 3 (Convex Function)**.** A function f is convex if the area above the function, $\text{epi}(f)$, is a convex set, where

$$\text{epi}(f) = \{[x, \alpha]^\intercal \in \mathbb{R}^n \times \mathbb{R} : \alpha \geq f(x)\}$$

The setepi(f) is called the epigraph of f. The equivalent statement is $\forall x, y \in \mathbf{dom}(f)$, and $\theta \in [0, 1]$,

$$f(\theta x + (1-\theta)y) \leq \theta f(x) + (1-\theta)f(y)$$

Theorem 1 (First-Order and Second-Order Convexity Condition)**.** Suppose f is twice-differentiable over an open domain. The the following are equivalent:

1. f is convex;
2. 1st-order condition:

$$f(y) \geq f(x) + \nabla f(x)^\intercal (y - x) \tag{1}$$

3. 2nd-order condition: the Hessian matrix of f is positive semidenite, i.e., the derivative is nondecreasing.

Proof. For the sake of simplicity, we only prove the result in case $n = 1$:

- 1. $\Rightarrow$ 2.: Suppose f is convex and $x, y, \in \mathbf{dom}(f)$. Since $\mathbf{dom}(f)$ is convex, $\forall \theta \in (0, 1], (1-\theta)x + \theta y = x + \theta(y - x) \in \mathbf{dom}(f)$. By convexity,

$$f(x + \theta(y - x)) \leq (1-\theta)f(x) + \theta f(y) \tag{2}$$

  Rearranging (2) yields

$$\begin{aligned} f(y) &\geq f(x) + \frac{f(x + \theta(y-x)) - f(x)}{\theta} \\ &= f(x) + (y - x) \times \frac{f(x + \theta(y-x)) - f(x)}{\theta(y-x)} \\ &\to f(x) + (y - x) \times f'(x) \text{ as } \theta \to 0 \end{aligned}$$

- 1. $\Leftarrow$ 2.: Suppose f statisfies (1). $\forall x, y \in \mathbf{dom}(f), x \neq y$ and $\theta \in [0,1]$, setting $z = \theta x + (1-\theta)y$, we have

$$f(x) \geq f(z) + f'(z)(x-z)$$

and

$$f(y) \geq f(z) + f'(z)(y-z)$$

Multiplying the first inequality by θ, the second by $1-\theta$, and adding them yields

$$\begin{aligned}\theta f(x) + (1-\theta)f(y) &\geq \theta\big(f(z) + f'(z)(x-z)\big) + (1-\theta)\big(f(z) + f'(z)(y-z)\big) \\ &= \theta\big(f(z) + f'(z)(1-\theta)(x-y)\big) + (1-\theta)\big(f(z) + f'(z)\theta(x-y)\big) \\ &= f(z)\end{aligned}$$

- 2. $\Rightarrow$ 3.: $\forall x, y \in \mathbf{dom}(f), y > x$, we have

$$f(y) \geq f(x) + f'(x)(y-x) \text{ and } f(x) \geq f(y) + f'(y)(x-y)$$

hence

$$f'(x)(y-x) \leq f(y) - f(x) \leq f'(y)(y-x) \tag{3}$$

Rearranging (3) obtains

$$\frac{f'(y) - f'(x)}{y-x} \geq 0$$

As we let $y \to x$, we get

$$f''(x) \geq 0, \forall x \in \mathbf{dom}(f)$$

- 2. $\Leftarrow$ 3.: Suppose $f''(x) \geq 0, \forall x \in \mathbf{dom}(f)$. By the mean value version of Taylors theorem, $\forall x, y \in \mathbf{dom}(f), x < y$, $\exists z \in [x, y]$ s.t.

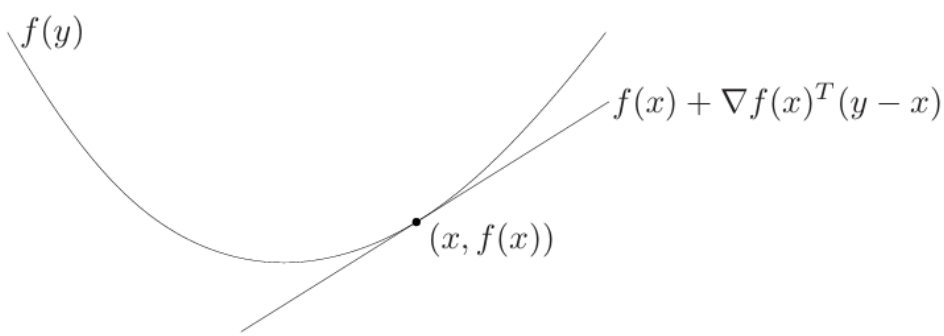
$$f(y) = f(x) + f'(x)(y-x) + f''(z)(y-x)^2$$

Since $f''(x)(y-x)^2 \geq 0$, we obtain

$$f(y) \geq f(x) + f'(x)(y-x)$$

□

The inequality (1) states that for a convex function, the first-order Taylor approximation is in fact a global underestimator of the function. Conversely, if the first-order Taylor approximation of a function is always a global underestimator of the function, then the function is convex.



Example 1 (Univariate Convex Function)**.**

- Linear:

$$f(x) = a + bx$$

- Quadratic:

$$f(x) = a + bx + cx^2$$

- Exponential:

$$f(x) = e^{ax}$$

- Negative logarithm

$$f(x) = -\log x$$

Example 2 (Multivariate Convex Function)**.**

- Affine function:

$$f(x) = a^\intercal x + b$$

- **Any norm**: Recall the definition of norm: Given a vector space $\mathbf{X}$ over a subfield $\mathbf{F}$ of the complex number $\mathbb{C}$, a norm on $\mathbf{X}$ is a real-valued function $\|\cdot\| : \mathbf{X} \to \mathbb{R}$ with the following properties:

  1. Subadditivity/Triangle inequality: $\|x + y\| \leq \|x\| + \|y\|, \forall x, y \in \mathbf{X}$;
  2. Absolute homogeneity: $\|\alpha x\| = |\alpha|\|x\|, \forall x \in \mathbf{X}, \alpha \in \mathbb{R}$;
  3. Positive definiteness/positiveness/Point-separating: if $\forall x \in \mathbf{X}, \|x\| = 0 \Rightarrow x = 0$.

  Applying properties 1. and 2. yields

$$\|\theta x + (1 - \theta y)\| \leq \|\theta x\| + \|(1 - \theta)y\| = \theta\|x\| + (1 - \theta)\|y\|, \forall x, y \in \mathbf{X}, \theta \in [0, 1]$$

Note: In this handout, we set $\|\cdot\|$ to be the Euclidean norm, i.e., $\|x\| = \sqrt{\sum_{i=1}^n x_i^2}$.

Definition 4 (Strict and Strong Convexity)**.** A function $f : \mathbb{R}^n \to \mathbb{R}$ is

- **strictly convex**, if $\forall x, y \in \mathbf{dom}(f), x \neq y$, and $\theta \in (0, 1)$,

$$f(\theta x + (1 - \theta)y) < \theta f(x) + (1 - \theta)f(y)$$

- m**-strongly convex**, if $\exists m > 0$ s.t. $f(x) - m\|x\|^2$ is convex.

Lemma 1. Strong convexity $\Rightarrow$ Strict convexity $\Rightarrow$ Convexity

Proof. Since strict convexity $\Rightarrow$ convexity is obvious, we only prove strong convexity $\Rightarrow$ strict convexity. If a fucntion f is strong convexity, $\forall x, y \in \mathbf{dom}(f), x \neq y$, and $\theta \in (0, 1)$, we have

$$f(\theta x + (1 - \theta y)) - m\|\theta x + (1 - \theta y)\|^2 \leq \theta f(x) + (1 - \theta)f(y) - \theta m\|x\|^2 - (1 - \theta)m\|y\|^2$$

Since $\|.\|^2$ is a strict convex function

$$-\theta m\|x\|^2 - (1-\theta)m\|y\|^2 < -m\|\theta x + (1-\theta y)\|^2$$

and hence

$$f(\theta x + (1-\theta y)) - m\|\theta x + (1-\theta y)\|^2 < \theta f(x) + (1-\theta)f(y) - \theta m\|x\|^2 - (1-\theta)m\|y\|^2$$

□

Note that if f is a m-strongly convex function, the function $g(x) = f(x) - m\|x\|^2$ is a convex function. By the first-order condition of convex function, we have

$$\begin{aligned} & g(y) \geq g(x) + \nabla g(x)^\intercal (y-x) \\ \Rightarrow\ & f(y) - m\|y\|^2 \geq f(x) - m\|x\|^2 + (\nabla f(x) - 2mx)^\intercal (y-x) \\ \Rightarrow\ & f(y) \geq f(x) + \nabla f(x)^\intercal (y-x) + m\|y-x\|^2 \end{aligned} \tag{4}$$

Gradient Descent Methods

The materials is taken from

- Audet and Hare (2017) chapter 2;
- Boyd and Vandenberghe (2004) chapter 9;
- Balakrishnan (2023) lecture 3;
- Schmidt (2018) lecture 4;
- Wang (2019) lecture 3.

Condiser an unconstrained optimization problem

$$\min_{x \in \mathbb{R}^n} f(x)$$

where $f : \mathbb{R}^n \to \mathbb{R}$ is a twice-differentiable convex function. The algorithms described in this note produce a minimizing sequence $x_k, k = 1, \ldots,$ where

$$x_{k+1} = x_k + t_k \Delta x_k$$

and $t_k > 0$. We call Δx_k a **descent direction** which is defined as follows:

Definition 5 (Descent Direction)**.** A direction $\Delta x_k \in \mathbb{R}^n$ is said to be a descent direction of the function f at the point $x_k \in \mathbb{R}^n$ if and only if there exists a scalar $t_k \in (0, \bar{t}]$ such that

$$f(x + t_k \Delta x_k) < f(x) \forall t_k \in (0, \bar{t}]$$

The outline of descent method is written as follows

1: General descent method

input: $f, x \in \mathbb{R}^n, k \leftarrow 0$

do

1: Determine a descent direction Δx_k

2: *Line search*: Choose a step size t_k

3: *Update*: $x_{k+1} = x_k + t_k \Delta x_k$

4: $k \leftarrow k + 1$

until (stopping criterion is satisfied)

The first step is searching the descent direction, from (1) we know that $f(x_{k+1}) = f(x_k + t_k \Delta x_k) \geq f(x_k)$ implies

$$\nabla f(x)^\intercal t_k \Delta x_k \geq 0$$

Since $t_k \geq 0$, the descent direction must satisfy

$$\nabla f(x)^\intercal \Delta x_k < 0$$

Therefore, the natural choice of Δx_k is $-\nabla f(x)$. This method is called **gradient descent method**.

The second question is how to choose the step size t_k? Here are som posibilities:

- Fixed step size
- Exact line search
- Inexact line search

Fixed Step Size

We assume ∇f is L-Lipschitz i.e. $\exists L > 0$ s.t.

$$\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|, \forall x, y$$

Theorem 2 (Quadratic Upper Bound)**.** ∇f is L-Lipschitz implies

1. $$(\nabla f(x) - \nabla f(y))^\intercal (x - y) \leq L\|x - y\|^2, \forall x, y \in \mathbf{dom}(f) \tag{5}$$

2. If $\mathbf{dom}(f)$ is convex, (5) is equivalent to

$$f(y) \leq f(x) + f(x)^\intercal (y - x) + \frac{L}{2}\|y - x\|^2, \forall x, y \in \mathbf{dom}(f)$$

We state the descent lemma of gradient descent algorithm

Lemma 2 (Descent Lemma)**.** If we run gradient descent for $k + 1$ iterations with a fixed step-size of $\frac{1}{L}$, it will yield a solution $f(x_{k+1})$ which satisfies

$$f(x_{k+1}) \leq f(x_k) - \frac{1}{2L}\|\nabla f(x_k)\|^2$$

Proof. The gradient descent update $x_{k+1} = x_k - \frac{1}{L}\nabla f(x_k)$ gives

$$x_{k+1} - x_k = -\frac{1}{L}\nabla f(x_k)$$

Since ∇f is L-Lipschitz, applying the quadratic bound property yields

$$\begin{aligned}
f(x_{k+1}) &\leq f(x_k) + \nabla f(x_k)^\intercal (x_{k+1} - x_k) + \frac{L}{2}\|x_{k+1} - x_k\|^2 \\
&= f(x_k) - \nabla f(x_k)^\intercal \left(\frac{1}{L}\nabla f(x_k)\right) + \frac{L}{2}\left\|\frac{1}{L}\nabla f(x_k)\right\|^2 \\
&= f(x_k) - \frac{L}{2}\|\nabla f(x_k)\|^2
\end{aligned}$$

□

According to Lemma 2, we start at some $f(x_0)$, and at each step we decrease f by at least $\frac{L}{2}\|\nabla f(x_k)\|^2$. So $\|\nabla f(x_k)\|^2$ must be going to zero **fast enough**. Denote x^* as an optimal solution for f, and f^* as the optimal value. We have the following convergence guarantee for convex L-smooth function f:

Theorem 3. Gradient descent with fxed step size $t \leq \frac{1}{L}$ satisfies

$$f(x_k) - f^* \leq \frac{\|x_0 - x^*\|^2}{2tk}$$

Proof. Since f is a convex function, we have

$$f(x_k) \leq f^* + \nabla f(x_k)^\intercal (x_k - x^*)$$

Combined with Lemma 2, we have:

$$\begin{aligned}
f(x_{k+1}) &= f(x_k) - \frac{t}{2}\|\nabla f(x_k)\|^2 \\
&\leq f^* + \nabla f(x_k)^\intercal (x_k - x^*) - \frac{t}{2}\|\nabla f(x_k)\|^2 \\
&= f^* - \frac{1}{2t}\left(\|t\nabla f(x_k)\|^2 - 2t\nabla f(x_k)^\intercal (x_k - x^*) + \|x_k - x^*\|^2 - \|x_k - x^*\|^2\right) \\
&= f^* - \frac{1}{2t}\left(\|t\nabla f(x_k) - (x_k - x^*)\|^2 - \|x_k - x^*\|^2\right) \\
&= f^* - \frac{1}{2t}\left(\|x_{k+1} - x^*\|^2 - \|x_k - x^*\|^2\right)
\end{aligned}$$

Telescoping from 1 to k, we can conclude the proof:

$$
\begin{aligned}
&\sum_{i=1}^{k} f(x_i) - kf^* \leq -\frac{1}{2t}\big(\|x_k - x^*\|^2 - \|x_0 - x^*\|^2\big) \\
\Rightarrow\ &\frac{1}{k}\sum_{i=1}^{k} f(x_i) - f^* \leq \frac{\|x_0 - x^*\|^2}{2tk} \\
\Rightarrow\ &f(x_k) - f^* \leq \frac{\|x_0 - x^*\|^2}{2tk}
\end{aligned}
$$

□

Exact Line Search

The correct step size in many cases may depend on properties of the function that we don't know. In practice, L is usually expensive to compute, and $t_k = 1/L$ is really small, hence *we only need a step size this small in the worst case*. We consider solving the following 1D optimization problem to determine the best step-size

$$t_k = \min_{\tilde{t}>0} f(x_k - \tilde{t}\nabla f(x_k))$$

Its often computationally cumbersome to solve this optimization problem exactly, so we resort to some approximation of this idea.

Inexact Line Search

Most of practitioners use **inexact line search** - that is to say, the step size is chosen to just reduce f *enough*. The simplest inexact line search is **backtracking line search**.

2: Backtracking line search

input: $x, \Delta x \in \mathbb{R}^n, \alpha \in (0, 0.5), \beta \in (0, 1), \text{maxIter} \in \mathbb{N}, k \leftarrow 0, t \leftarrow 1$

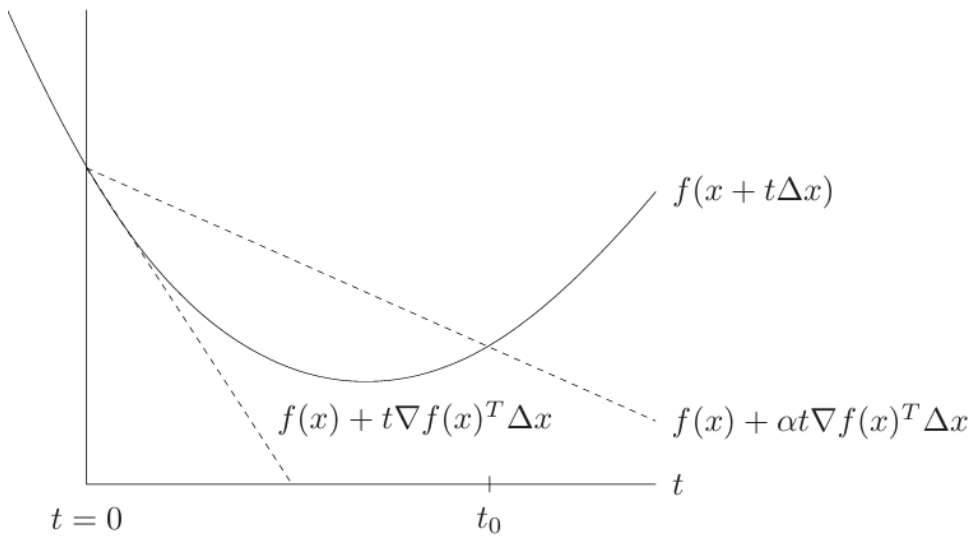
do while ($k < \text{maxIter}$ and $f(x + t\Delta x) > f(x) + \alpha t \nabla f(x)^\intercal \Delta x$)

1: $k \leftarrow k + 1$

2: $t \leftarrow \beta t$

Since Δx is a descent direction, $\nabla f(x)^\intercal \Delta x < 0$. For small enough t, we have

$$f(x + t\Delta x) \approx f(x) + t\nabla f(x)^\intercal \Delta x < f(x) + \alpha t \nabla f(x)^\intercal \Delta x$$



Theorem 4. Suppose the function$f : \mathbb{R}^n \to \mathbb{R}$ is convex and differentiable, and ∇f is L-Lipschitz. Then if we run gradient descent for k iterations with step size t_k chosen using backtracking line search on each iteration k, it will yield a solution $f(x_k)$ which satisfies

$$f(x_k) - f^* \leq \frac{\|x_0 - x^*\|}{2t_{\min} k}$$

where $t_{\min} = \min\{1, \beta/L\}$.

Notice that if we choose β to be large enough, the rate of convergence is very similar to what we got for gradient descent with fixed step size.

The Strongly Convex Case

Lemma 3 (A Gradient Bound under Smoothness)**.**

$$\|\nabla f(x)\|^2 \leq 2L(f(x) - f^*)$$

Proof. By quadratic upper bound, we have

$$f^* \leq f\left(x - \frac{1}{L}\nabla f(x)\right) \leq f(x) - \frac{1}{L}\|\nabla f(x)\|^2 + \frac{1}{2L}\|\nabla f(x)\|^2 = f(x) - \frac{1}{2L}\|\nabla f(x)\|^2$$

Rearranging above equation yields

$$\|\nabla f(x)\|^2 \leq 2L(f(x) - f^*)$$

□

Theorem 5. Let f be a m-strongly convex and L-smooth function. Gradient descent with step size $t = 1/L$, we have

$$\|x_k - x^*\| \leq \left(1 - \frac{2m}{L}\right)^k \|x_0 - x^*\|$$

Proof.

$$\begin{aligned}
\|x_{k+1} - x^*\|^2 &= \|x_k - \nabla f(x) - x^*\|^2 \\
&= \|x_k - x^*\|^2 - 2t\nabla f(x_k)^\intercal(x_k - x^*) + t^2\|\nabla f(x_k)\|^2 \\
&\leq \|x_k - x^*\|^2 - 2t(f(x_k) - f^* + m\|x_k - x^*\|^2) + t^2\|\nabla f(x_k)\|^2 \quad \text{(By (4))} \\
&\leq \|x_k - x^*\|^2 - 2t(f(x_k) - f^* + m\|x_k - x^*\|^2) + 2t^2L(f(x_k) - f^*) \quad \text{(By Lemma 3)} \\
&= \|x_k - x^*\|^2 - 2tm\|x_k - x^*\|^2 + 2t(tL - 1)(f(x_k) - f^*) \\
&\leq \|x_k - x^*\|^2 - 2tm\|x_k - x^*\|^2 \\
&= \left(1 - \frac{2m}{L}\right)\|x_k - x^*\|
\end{aligned}$$

□

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